

Introduction to Big Data, Spring 2026

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Assignment 2

Methodology:

My search engine pipeline consists of the the following stages:

- 1) **Data preparation.** At this stage, a parquet file (I used e.parquet, but renamed it to a.parquet to not change template repository code) is read from HDFS using PySpark DataFrame. PySpark DataFrame selects only the <id>, <title>, and <text> fields from a row that represents a Wikipedia article, filters out rows where the text is null, and limits the output to 1000 documents. For each selected row, a text file is created in the local /data folder, with the filename formatted as <id>_<title>.txt. These files are then uploaded to HDFS /data folder.

After it, the HDFS /data folder is used as input for a PySpark RDD, which collects all the individual files and transforms them into a single partition. Each line in this partition is formatted as <id>\t<title>\t<text>. The result is stored in HDFS at /input/data and serves as the input for the MapReduce indexing pipeline.

- 2) **Indexing in HDFS.** At this stage, 4 MapReduce pipelines are performed on /input/data single partition. I chose this number of pipelines because each reducer outputs a single file, and I needed four separate outputs: vocabulary, inverted index, document statistics, and corpus statistics. All of these components are necessary for BM25 calculation, and storing them in separate locations makes the data easier to manage and access during the search phase:

- The first pipeline uses mapper1.py and identity reducer. The goal of it is to extract document length and store it with document ID and title for convenient query output in the future. Mapper reads each line of the input/data partition, tokenizes text (uses lowercase and cleans from all punctuation characters), counts text length and prints the output in the format <doc_id>\t<title>\t<text_length>. Identity reducer passes the mapper's output unchanged. The result is stored in /indexer/doc_stats in HDFS, where each line contains a document's ID, title, and word count.
- The second pipeline uses mapper2.py and reducer2.py. The goal of it is to count the total number of documents in input/data and compute their average text length. Mapper just reads the output from

/indexer/doc_stats and passes it unchanged to reducer. Reducer counts the total number of documents and sums their lengths, then computes the average document length. The result is stored in the /indexer/corpus_stats in HDFS, which consists of only one line <total_doc_number>\t<avg_text_length>.

- The third pipeline uses mapper3.py and reducer3.py. The goal of it is to compute inverted indexes of document terms. Mapper reads each line of the input/data partition, tokenizes text (uses lowercase and cleans from all punctuation characters), counts the frequency of each unique term in the document, and prints the output in the format <term>\t<doc_id>\t<term_frequency>. During the shuffle phase, Hadoop automatically groups all records by term. Reducer receives each term once, along with all associated <doc_id>, <tf> pairs, and builds a posting list in the format <doc_id_1>:<tf_1>,<doc_id_2>:<tf_2>,.... The result is stored in /indexer/term_doc_counts in HDFS, where each line contains <term>\t<postings>.
- The fourth pipeline uses mapper4.py and reducer4.py. The goal of it is to build vocabulary of unique terms. Mapper just reads the output from /indexer/term_doc_counts and passes it unchanged to reducer. Reducer receives it in the format <term>\t<postings> and splits posting string by “,” to count the number of documents a term is present in (document frequency). The result is stored in /indexer/vocabulary in HDFS, where each line contains <term>\t<df>.

3) Storing HDFS index data to ScyllaDB. At this stage, I connect to ScyllaDB using cassandra-driver since it's easier to use on the small amount of data than PySpark Connector. Then, I create keyspace “search_engine” and define there the following 4 tables (for every MapReduce output):

- **vocabulary table** takes data stored in indexer/vocabulary and includes *term (text)* as primary key and *df (int)* as attribute.
- **inverted_index table** takes data stored in indexer/term_doc_counts and includes *term (text)* as primary key and *postings (text)* as attribute.
- **documents_stats table** takes data stored in indexer/doc_stats and includes *doc_id (text)* as primary key and *title (text)*, *doc_length (int)* as attribute.
- **corpus_stats table** takes data stored in indexer/corpus_stats and includes *id = 1* as primary key and *total_docs (int)*, *avg_doc_length (float)* as attribute. Cassandra requires every table to have a primary key. Since corpus_stats has no natural unique identifier, artificial id = 1 is used to ensure the table can store exactly one row.

4) Querying and getting ranker's top-10 documents based on BM25 metric. On this stage, querying is performed by a PySpark application that runs on

YARN cluster via search.sh. The script passes the user's query as an argument of search.sh, activates the virtual environment, and uses spark-submit with the Cassandra connector and archived dependencies to run query.py on the cluster using input from ScyllaDB.

query.py performs BM25 calculation and ranking. First, it connects to ScyllaDB from PySpark and reads the four tables (vocabulary, inverted_index, document_stats, and corpus_stats) into PySpark DataFrames: vocab_df, inverted_df, doc_stats_df, and corpus_df. I chose DataFrames over RDDs because they provide a higher-level, optimized interface that is more convenient when working with structured tables requiring joins and aggregations beyond simple transformations.

Second, it extracts necessary data from created DataFrames:

- N and avg_dl from corpus_df
- relevant_vocab from vocab_df filtered by tokenized unique query terms (tokenization uses the same logic as in the MapReduce pipelines: lowercasing and punctuation removal)
- query_inverted from inverted_df joined with relevant_vocab on *term*
- exploded_df from parsing the posting strings in query_inverted (to do it, I used udf (user defined function) that splits each string by “,” to isolate individual posting entries, then by “:” to extract *doc_id* and *tf*, returning an array of tuples (*doc_id*, *int(tf)*) for each term. The result is then exploded, expanding each (*term*, *doc_id*, *tf*) combination into a separate row. This transformation enables the data to be joined with document statistics for BM25 calculation)
- with_doc_stats from exploded_df joined with doc_stats_df on *doc_id*. Therefore, at the end with_doc_stats has the structure: term (t), df, doc_id (d), tf, title, doc_length.

Third, with_doc_stats is used for BM25 calculation, which follows the formula:

$$BM25(t, d) = \log\left(\frac{N}{df(t)}\right) \times \frac{(k_1 + 1) \times tf(t, d)}{k_1 \times (1 - b + b \times \frac{doc_length(d)}{avg_dl}) + tf(t, d)}$$

where N is total documents, df(t) is document frequency, tf(t,d) is term frequency in document, doc_length(d) is document length, avg_dl is average document length, and k1 = 1.0, b = 0.75 are standard BM25 parameters.

The BM25 score is added as a new column to with_doc_stats using withColumn function, where the score parameter is a column expression that combines the IDF and normalized term frequency components using PySpark SQL functions (log, lit, col).

Finally, the results are generated by grouping the DataFrame by *doc_id* and *title*, summing the BM25 scores for each document to obtain a total relevance

score. The documents are then ordered by this total score in descending order, and the top 10 are selected. They are printed as the final search output, displaying each document's ID, title, and total score.

Demonstration:

Guide on how to run the repository:

To run the code, clone the GitHub repository

(https://github.com/arinapetukhova/big_data_assignment2), place the a.parquet file with Wikipedia data inside the app/ folder of the cloned repository, then run *docker compose up* from the repository root folder. This will start the necessary containers (Spark master node, Spark slave node, ScyllaDB) and automatically execute app/app.sh on the Spark master node, which starts all necessary services (HDFS, YARN), creates virtual environment, installs on it needed packages, and runs the complete search engine pipeline. To change the query or add more queries, modify the argument of search.sh in app/app.sh, or add additional lines with *bash search.sh "query text"* to run multiple searches. To index more text files, change n = 1000 in prepare_data.py.

Screenshots of successful indexing of the documents:

```
cluster-master | Creating index in HDFS...
cluster-master | Building document statistics...
cluster-master | packageJobJar: [/tmp/hadoop-unjar1238317353828858782/] [] /tmp/streamjob9095767991341238332.jar tmpDir=null
cluster-master | 2026-03-23 12:26:36,088 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
cluster-master | 2026-03-23 12:26:36,087 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
cluster-master | 2026-03-23 12:26:37,291 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1774268073042_0001
cluster-master | 2026-03-23 12:26:38,162 INFO mapred.FileInputFormat: Total input files to process : 1
cluster-master | 2026-03-23 12:26:38,468 INFO mapreduce.JobSubmitter: number of splits:2
cluster-master | 2026-03-23 12:26:39,516 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774268073042_0001
cluster-master | 2026-03-23 12:26:39,516 INFO mapreduce.JobSubmitter: Executing with tokens: []
cluster-master | 2026-03-23 12:26:39,897 INFO conf.Configuration: resource-types.xml not found
cluster-master | 2026-03-23 12:26:39,897 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
cluster-master | 2026-03-23 12:26:40,498 INFO impl.YarnClientImpl: Submitted application application_1774268073042_0001
cluster-master | 2026-03-23 12:26:40,557 INFO mapreduce.Job: The url to track the job: http://cluster-master:8080/proxy/application_1774268073042_0001/
cluster-master | 2026-03-23 12:26:40,559 INFO mapreduce.Job: Running job: job_1774268073042_0001
cluster-master | 2026-03-23 12:26:51,728 INFO mapreduce.Job: Job job_1774268073042_0001 running in uber mode : false
cluster-master | 2026-03-23 12:26:51,722 INFO mapreduce.Job: map 0% reduce 0%
cluster-master | 2026-03-23 12:26:58,838 INFO mapreduce.Job: map 100% reduce 0%
cluster-master | 2026-03-23 12:27:14,924 INFO mapreduce.Job: map 100% reduce 100%
cluster-master | 2026-03-23 12:27:15,943 INFO mapreduce.Job: Job job_1774268073042_0001 completed successfully
cluster-master | 2026-03-23 12:27:16,122 INFO mapreduce.Job: Counters: 64
cluster-master |
cluster-master | File System Counters
cluster-master |   FILE: Number of bytes read=67152
cluster-master |   FILE: Number of bytes written=62957
cluster-master |   FILE: Number of read operations=0
cluster-master |   FILE: Number of large read operations=0
cluster-master |   FILE: Number of write operations=0
cluster-master |   HDFS: Number of bytes read=866785
cluster-master |   HDFS: Number of bytes written=63216
cluster-master |   HDFS: Number of read operations=11
cluster-master |   HDFS: Number of large read operations=0
cluster-master |   HDFS: Number of write operations=2
cluster-master |   HDFS: Number of bytes read erasure-coded=0
cluster-master |
cluster-master | Job Counters
cluster-master |   Launched map tasks=2
cluster-master |   Launched reduce tasks=1
cluster-master |   Data-local map tasks=2
cluster-master |   Total time spent by all maps in occupied slots (ms)=18044
cluster-master |   Total time spent by all reduces in occupied slots (ms)=12685
cluster-master |   Total time spent by all map tasks (ms)=18044
cluster-master |   Total time spent by all reduce tasks (ms)=12685
cluster-master |   Total vcore-milliseconds taken by all map tasks=18044
cluster-master |   Total vcore-milliseconds taken by all reduce tasks=12685
cluster-master |   Total megabyte-milliseconds taken by all map tasks=18285856
cluster-master |   Total megabyte-milliseconds taken by all reduce tasks=12987528
cluster-master |
cluster-master | Map-Reduce Framework
cluster-master |   Map input records=1965
cluster-master |   Map output records=1965
cluster-master |   Map output bytes=63216
cluster-master |   Map output materialized bytes=67158
cluster-master |   Input split bytes=200
cluster-master |   Combine input records=0
cluster-master |   Combine output records=0
cluster-master |   Reduce input groups=1965
cluster-master |   Reduce shuffle bytes=67158
cluster-master |   Reduce input records=1965
cluster-master |   Reduce output records=1965
cluster-master |   Spilled Records=3930
cluster-master |   Shuffled Maps =2
cluster-master |   Failed Shuffles=0
cluster-master |   Merged Map outputs=2
cluster-master |   GC time elapsed (ms)=229
cluster-master |   CPU time spent (ms)=2880
cluster-master |   Physical memory (bytes) snapshot=788644864
cluster-master |   Virtual memory (bytes) snapshot=7626891264
cluster-master |   Total committed heap usage (bytes)=580386816
cluster-master |   Peak Map Physical memory (bytes)=278654976
```



```
cluster-master | Peak Map Virtual memory (bytes)=2543345664
cluster-master | Peak Reduce Physical memory (bytes)=161845248
cluster-master | Peak Reduce Virtual memory (bytes)=2545494944
cluster-master | Shuffle
cluster-master |   Errors
cluster-master |     BAD_ID=0
cluster-master |     CONNECTION=0
cluster-master |     IO_ERROR=0
cluster-master |     WRONG_LENGTH=0
cluster-master |     WRONG_MAP=0
cluster-master |     WRONG_REDUCE=0
cluster-master | File Input Format Counters
cluster-master |   Bytes Read=8866585
cluster-master | File Output Format Counters
cluster-master |   Bytes Written=63216
cluster-master | 2026-03-23 12:27:16,122 INFO streaming.StreamJob: Output directory: /indexer/doc_stats
cluster-master | Building corpus statistics...
cluster-master | packageJobJar: [/tmp/hadoop-unjar135521494368753558/] [] /tmp/streamjob5748818562164526747.jar tmpDir=null
cluster-master | 2026-03-23 12:27:18,776 INFO client.DefaultHadoopFollowerProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
cluster-master | 2026-03-23 12:27:19,436 INFO client.DefaultHadoopFollowerProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
cluster-master | 2026-03-23 12:27:19,985 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1774268073842_0002
cluster-master | 2026-03-23 12:27:20,575 INFO mapred.FileInputFormat: Total input files to process : 1
cluster-master | 2026-03-23 12:27:20,849 INFO mapreduce.JobSubmitter: number of splits:2
cluster-master | 2026-03-23 12:27:21,168 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774268073842_0002
cluster-master | 2026-03-23 12:27:21,168 INFO mapreduce.JobSubmitter: Executing with tokens: []
cluster-master | 2026-03-23 12:27:21,518 INFO conf.Configuration: resource-types.xml not found
cluster-master | 2026-03-23 12:27:21,519 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
cluster-master | 2026-03-23 12:27:21,618 INFO impl.YarnClientImpl: Submitted application application_1774268073842_0002
cluster-master | 2026-03-23 12:27:21,663 INFO mapreduce.Job: The url to track the job: http://cluster-master:8088/proxy/application_1774268073842_0002/
cluster-master | 2026-03-23 12:27:21,664 INFO mapreduce.Job: Running job: job_1774268073842_0002
cluster-master | 2026-03-23 12:27:30,776 INFO mapreduce.Job: Job job_1774268073842_0002 running in uber mode : false
cluster-master | 2026-03-23 12:27:30,778 INFO mapreduce.Job: map 0% reduce 0%
cluster-master | 2026-03-23 12:27:35,364 INFO mapreduce.Job: map 100% reduce 0%
cluster-master | 2026-03-23 12:27:41,876 INFO mapreduce.Job: map 100% reduce 100%
cluster-master | 2026-03-23 12:27:42,889 INFO mapreduce.Job: Job job_1774268073842_0002 completed successfully
cluster-master | 2026-03-23 12:27:42,790 INFO mapreduce.Job: Counters: 64
cluster-master | File System Counters
cluster-master |   FILE: Number of bytes read=67152
cluster-master |   FILE: Number of bytes written=9351
cluster-master |   FILE: Number of read operations=0
cluster-master |   FILE: Number of large read operations=0
cluster-master |   FILE: Number of write operations=0
cluster-master |   HDFS: Number of bytes read=67526
cluster-master |   HDFS: Number of bytes written=23
cluster-master |   HDFS: Number of read operations=11
cluster-master |   HDFS: Number of large read operations=0
cluster-master |   HDFS: Number of write operations=2
cluster-master |   HDFS: Number of bytes read erasure-coded=0
cluster-master | Job Counters
cluster-master |   Launched map tasks=2
cluster-master |   Launched reduce tasks=1
cluster-master |   Data-local map tasks=2
cluster-master |   Total time spent by all maps in occupied slots (ms)=6855
cluster-master |   Total time spent by all reduces in occupied slots (ms)=2858
cluster-master |   Total time spent by all map tasks (ms)=4865
cluster-master |   Total time spent by all reduce tasks (ms)=2858
cluster-master |   Total vcore-millisecods taken by all map tasks=6055
cluster-master |   Total vcore-millisecods taken by all reduce tasks=2858
cluster-master |   Total megabyte-millisecods taken by all map tasks=6208320
cluster-master |   Total megabyte-millisecods taken by all reduce tasks=2918400
cluster-master | Map-Reduce Framework
cluster-master |   Map input records=1965
cluster-master |   Map output records=1965
cluster-master |   Map output bytes=63216
cluster-master |   Map output materialized bytes=67158
cluster-master |   Input split bytes=214
cluster-master |   Combine input records=0
cluster-master |   Combine output records=0
cluster-master |
cluster-master | Reduce input groups=1965
cluster-master | Reduce shuffle bytes=67158
cluster-master | Reduce input records=1965
cluster-master | Reduce output records=1
cluster-master | Spilled Records=9390
cluster-master | Shuffled Maps =2
cluster-master | Failed Shuffles=0
cluster-master | Merged Map outputs=2
cluster-master | GC time elapsed (ms)=253
cluster-master | CPU time spent (ms)=1908
cluster-master | Physical memory (bytes) snapshot=770191360
cluster-master | Virtual memory (bytes) snapshot=7628853248
cluster-master | Total committed heap usage (bytes)=623902720
cluster-master | Peak Map Physical memory (bytes)=292892672
cluster-master | Peak Map Virtual memory (bytes)=2543247360
cluster-master | Peak Reduce Physical memory (bytes)=192897024
cluster-master | Peak Reduce Virtual memory (bytes)=2546622464
cluster-master | Shuffle Errors
cluster-master |   BAD_ID=0
cluster-master |   CONNECTION=0
cluster-master |   IO_ERROR=0
cluster-master |   WRONG_LENGTH=0
cluster-master |   WRONG_MAP=0
cluster-master |   WRONG_REDUCE=0
cluster-master | File Input Format Counters
cluster-master |   Bytes Read=47312
cluster-master | File Output Format Counters
cluster-master |   Bytes Written=23
cluster-master | 2026-03-23 12:27:42,978 INFO streaming.StreamJob: Output directory: /indexer/corpus_stats
cluster-master | Building inverted index...
cluster-master | packageJobJar: [/tmp/hadoop-unjar6395767353987758686/] [] /tmp/streamjob6448542069475798852.jar tmpDir=null
cluster-master | 2026-03-23 12:27:45,180 INFO client.DefaultHadoopFollowerProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
cluster-master | 2026-03-23 12:27:45,336 INFO client.DefaultHadoopFollowerProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
cluster-master | 2026-03-23 12:27:45,667 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1774268073842_0003
cluster-master | 2026-03-23 12:27:46,432 INFO mapred.FileInputFormat: Total input files to process : 1
cluster-master | 2026-03-23 12:27:46,725 INFO mapreduce.JobSubmitter: number of splits:2
cluster-master | 2026-03-23 12:27:46,963 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774268073842_0003
cluster-master | 2026-03-23 12:27:47,165 INFO mapreduce.JobSubmitter: Executing with tokens: []
cluster-master | 2026-03-23 12:27:47,166 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
cluster-master | 2026-03-23 12:27:47,238 INFO impl.YarnClientImpl: Submitted application application_1774268073842_0003
cluster-master | 2026-03-23 12:27:47,278 INFO mapreduce.Job: The url to track the job: http://cluster-master:8088/proxy/application_1774268073842_0003/
cluster-master | 2026-03-23 12:27:47,284 INFO mapreduce.Job: Running job: job_1774268073842_0003
cluster-master | 2026-03-23 12:27:54,385 INFO mapreduce.Job: Job job_1774268073842_0003 running in uber mode : false
cluster-master | 2026-03-23 12:27:54,389 INFO mapreduce.Job: map 0% reduce 0%
cluster-master | 2026-03-23 12:28:01,492 INFO mapreduce.Job: map 100% reduce 0%
cluster-master | 2026-03-23 12:28:07,535 INFO mapreduce.Job: map 100% reduce 100%
cluster-master | 2026-03-23 12:28:08,556 INFO mapreduce.Job: Job job_1774268073842_0003 completed successfully
cluster-master | 2026-03-23 12:28:08,662 INFO mapreduce.Job: Counters: 64
cluster-master | File System Counters
cluster-master |   FILE: Number of bytes read=10531991
cluster-master |   FILE: Number of bytes written=21893517
cluster-master |   FILE: Number of read operations=0
cluster-master |   FILE: Number of large read operations=0
cluster-master |   FILE: Number of write operations=0
cluster-master |   HDFS: Number of bytes read=8866785
cluster-master |   HDFS: Number of bytes written=621668
cluster-master |   HDFS: Number of read operations=11
cluster-master |   HDFS: Number of large read operations=0
cluster-master |   HDFS: Number of write operations=2
cluster-master |   HDFS: Number of bytes read erasure-coded=0
cluster-master | Job Counters
cluster-master |   Launched map tasks=2
cluster-master |   Launched reduce tasks=1
cluster-master |   Data-local map tasks=2
cluster-master |   Total time spent by all maps in occupied slots (ms)=8633
cluster-master |   Total time spent by all reduces in occupied slots (ms)=4853
```

```
cluster-master | Total time spent by all map tasks (ms)=8633
cluster-master | Total time spent by all reduce tasks (ms)=4053
cluster-master | Total vcore-milliseconds taken by all map tasks=8633
cluster-master | Total vcore-milliseconds taken by all reduce tasks=4053
cluster-master | Total megabyte-milliseconds taken by all map tasks=8840192
cluster-master | Total megabyte-milliseconds taken by all reduce tasks=4150272
cluster-master | Map-Reduce Framework
cluster-master |   Map input records=1965
cluster-master |   Map output records=532868
cluster-master |   Map output bytes=966248
cluster-master |   Map output materialized bytes=10531997
cluster-master |   Input split bytes=200
cluster-master |   Combine input records=0
cluster-master |   Combine output records=0
cluster-master |   Reduce input groups=82404
cluster-master |   Reduce shuffle bytes=10531997
cluster-master |   Reduce input records=532868
cluster-master |   Reduce output records=82404
cluster-master |   Spilled Records=105736
cluster-master |   Shuffled Maps =2
cluster-master |   Failed Shuffles=0
cluster-master |   Merged Map outputs=2
cluster-master |   GC time elapsed (ms)=201
cluster-master |   CPU time spent (ms)=6880
cluster-master |   Physical memory (bytes) snapshot=886158336
cluster-master |   Virtual memory (bytes) snapshot=7624589312
cluster-master |   Total committed heap usage (bytes)=672137216
cluster-master |   Peak Map Physical memory (bytes)=291287040
cluster-master |   Peak Map Virtual memory (bytes)=253959000
cluster-master |   Peak Reduce Physical memory (bytes)=225190800
cluster-master |   Peak Reduce Virtual memory (bytes)=254587206
cluster-master | Shuffle
cluster-master |   Errors
cluster-master |     BAD_ID=0
cluster-master |     CONNECTION=0
cluster-master |     ID_ERROR=0
cluster-master |     WRONG_LENGTH=0
cluster-master |     WRONG_MAP=0
cluster-master |     WRONG_REDUCE=0
cluster-master |   File Input Format Counters
cluster-master |     Bytes Read=8866505
cluster-master |   File Output Format Counters
cluster-master |     Bytes Written=8321868
cluster-master | 2026-03-23 12:28:08,664 INFO streaming.StreamJob: Output directory: /indexer/term_doc_counts
cluster-master | Building vocabulary...
cluster-master | packageJobJar: [/tmp/hadoop-unjar3216179967114620928/] [] /tmp/streamjob7185586249120562391.jar tmpDir=null
cluster-master | 2026-03-23 12:28:10,888 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
cluster-master | 2026-03-23 12:28:11,024 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at cluster-master/172.18.0.4:8032
cluster-master | 2026-03-23 12:28:11,320 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/root/.staging/job_1774268073842_0004
cluster-master | 2026-03-23 12:28:12,226 INFO mapred.FileInputFormat: Total input files to process : 1
cluster-master | 2026-03-23 12:28:12,551 INFO mapreduce.JobSubmitter: number of splits:2
cluster-master | 2026-03-23 12:28:12,874 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1774268073842_0004
cluster-master | 2026-03-23 12:28:12,874 INFO mapreduce.JobSubmitter: Executing with tokens: []
cluster-master | 2026-03-23 12:28:13,094 INFO conf.Configuration: resource-types.xml not found
cluster-master | 2026-03-23 12:28:13,094 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
cluster-master | 2026-03-23 12:28:13,168 INFO impl.YarnClientImpl: Submitted application application_1774268073842_0004
cluster-master | 2026-03-23 12:28:13,200 INFO mapreduce.Job: The url to track the job: http://cluster-master:8080/proxy/application_1774268073842_0004/
cluster-master | 2026-03-23 12:28:13,200 INFO mapreduce.Job: Job job_1774268073842_0004 running in uber mode : false
cluster-master | 2026-03-23 12:28:12,311 INFO mapreduce.Job: Job job_1774268073842_0004 completed successfully
cluster-master | 2026-03-23 12:28:35,590 INFO mapreduce.Job: Counters: 54
cluster-master |   File System Counters
cluster-master |     FILE: Number of bytes read=606490
cluster-master |     FILE: Number of bytes written=13842519
cluster-master |
cluster-master |   FILE: Number of read operations=0
cluster-master |   FILE: Number of large read operations=0
cluster-master |   FILE: Number of write operations=0
cluster-master |   HDFS: Number of bytes read=6326190
cluster-master |   HDFS: Number of bytes written=923431
cluster-master |   HDFS: Number of read operations=11
cluster-master |   HDFS: Number of large read operations=0
cluster-master |   HDFS: Number of write operations=2
cluster-master |   HDFS: Number of bytes read erasure-coded=0
cluster-master | Job Counters
cluster-master |   Launched map tasks=2
cluster-master |   Launched reduce tasks=1
cluster-master |   Deto-local map tasks=2
cluster-master |   Total time spent by all maps in occupied slots (ms)=7533
cluster-master |   Total time spent by all reduces in occupied slots (ms)=4744
cluster-master |   Total time spent by all map tasks (ms)=7533
cluster-master |   Total time spent by all reduce tasks (ms)=4744
cluster-master |   Total vcore-milliseconds taken by all map tasks=7533
cluster-master |   Total vcore-milliseconds taken by all reduce tasks=4744
cluster-master |   Total megabyte-milliseconds taken by all map tasks=7713792
cluster-master |   Total megabyte-milliseconds taken by all reduce tasks=4857856
cluster-master | Map-Reduce Framework
cluster-master |   Map input records=82404
cluster-master |   Map output records=82404
cluster-master |   Map output bytes=6331735
cluster-master |   Map output materialized bytes=6506486
cluster-master |   Input split bytes=226
cluster-master |   Combine input records=0
cluster-master |   Combine output records=0
cluster-master |   Reduce input groups=82404
cluster-master |   Reduce shuffle bytes=6506486
cluster-master |   Reduce input records=82404
cluster-master |   Reduce output records=82404
cluster-master |   Spilled Records=164088
cluster-master |   Shuffled Maps =2
cluster-master |   Failed Shuffles=0
cluster-master |   Merged Map outputs=2
cluster-master |   GC time elapsed (ms)=299
cluster-master |   CPU time spent (ms)=5050
cluster-master |   Physical memory (bytes) snapshot=794085084
cluster-master |   Virtual memory (bytes) snapshot=7629402112
cluster-master |   Total committed heap usage (bytes)=627572736
cluster-master |   Peak Map Physical memory (bytes)=293150720
cluster-master |   Peak Map Virtual memory (bytes)=254632032
cluster-master |   Peak Reduce Physical memory (bytes)=213901312
cluster-master |   Peak Reduce Virtual memory (bytes)=2547998720
cluster-master | Shuffle
cluster-master |   Errors
cluster-master |     BAD_ID=0
cluster-master |     CONNECTION=0
cluster-master |     ID_ERROR=0
cluster-master |     WRONG_LENGTH=0
cluster-master |     WRONG_MAP=0
cluster-master |     WRONG_REDUCE=0
cluster-master |   File Input Format Counters
cluster-master |     Bytes Read=6325964
cluster-master |   File Output Format Counters
cluster-master |     Bytes Written=923431
cluster-master | 2026-03-23 12:28:35,590 INFO streaming.StreamJob: Output directory: /indexer/vocabulary
cluster-master | create_index.sh completed
cluster-master | Storing index to ScyllaDB...
cluster-master | Creating schema...
scylladb-server | INFO 2026-03-23 12:28:39,746 [shard 0] migration_manager - Create new Keyspace: KSMetaData(name=search_engine, strategyClass=org.apache.cassandra.locator.SimpleStrategy, strategyOptions={replication_factor=1}, cfMeta
scylladb-server | Data=1, durable=1, userTypes=org.apache.cassandra.config.UTF8Metadata@60004a24540)
scylladb-server | INFO 2026-03-23 12:28:40,125 [shard 0] schema_tables - Creating keyspace search_engine
scylladb-server | INFO 2026-03-23 12:28:40,328 [shard 0] schema_tables - Schema version changed to b83c8199-493a-30b3-a628-c3e080b4b008
scylladb-server | INFO 2026-03-23 12:28:40,368 [shard 0] migration_manager - Create new ColumnFamily: org.apache.cassandra.config.CfMetaDataTable@20x600004127800[cfId=2d87a08-26a3-11f1-bcae-25eac9cfab1b,ksName=search_engine,cfName=vocabula
ry,cfType=Standard,comparator=org.apache.cassandra.db.marshall.CompositeType(org.apache.cassandra.db.marshall.UTF8Type),comment=,readRepairChance=0,dcLocalReadRepairChance=0,tombstoneGCOptions={\"mode\":\"timeout\", \"propagation_delay_in_secon
```


Screenshots of successful loading of data from HDFS to ScyllaDB:

```
cluster-master | create_index.sh completed...
cluster-master | Storing index to ScyllaDB...
cluster-master | Creating search engine
scylladb-server | INFO 2026-03-23 12:28:39,746 [shard 0] migration_manager - Create new Keyspace: K$Metadata(name=search_engine, strategyClass=org.apache.cassandra.locator.SimpleStrategy, strategyOptions={replication_factor=1}, c$Meta
Data={}, durable_writes=1, userType=org.apache.cassandra.config.UUIDMetadata@x600004ee1480)
scylladb-server | INFO 2026-03-23 12:28:41,395 [shard 0] schema_tables - Creating keyspace search_engine
scylladb-server | INFO 2026-03-23 12:28:40,320 [shard 0] schema_tables - Schema version changed to 883b199-493a-3803-a628-c3e88004b088
scylladb-server | INFO 2026-03-23 12:28:40,368 [shard 0] migration_manager - Create new ColumnFamily: org.apache.cassandra.config.CFMetadata@x600004127800[cfId=d2087a80-26b3-11f1-bcae-25eec9cfab1b, ksName=search_engine, cfName=vocabulary
Type=cfTypeStandard, comparator=org.apache.cassandra.db.marshall.Utf8Type, comment=, readRepairChance=0, dcLocalReadRepairChance=0, tombstoneOptions={mode=""}, timeout, propagation_delay_in_s
econds=""$0000}, gcTraceSeconds=864000, keyValidator=org.apache.cassandra.db.marshall.Utf8Type, minCompactionThreshold=4, maxCompactionThreshold=32, columnMetadata={ColumnDefinition(name=term, type=org.apache.cassandra.db.marshall.Utf8Type, kind=P
ARTITION_KEY, componentIndex=0, droppedAt=-9223372036854775808), ColumnDefinition(name=df, type=org.apache.cassandra.db.marshall.Int32Type, kind=REGULAR, componentIndex=null, droppedAt=-9223372036854775808)}, compactionStrategyClass=class
org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy, compactionStrategyOptions={stable_compression=org.apache.cassandra.io.compress.LZ4Compressor, bloomFilterChance=0.01, memtableFlushPeriod=0, cachingKeys="ALL", rows_per_partition="ALL", cdcId, defaultTimeToLive=0, minIndexInterval=128, maxIndexInterval=2048, speculativeRetry=99, @PERCENTILE, triggers=[]}, isDense=false, version=d89f0661-26b3-11f1-bcae-25eec9cfab1b, droppedColumns={}, collections={}, indices={}}
scylladb-server | INFO 2026-03-23 12:28:40,453 [shard 0] schema_tables - Creating search_engine.vocabulary id=d2087a80-26b3-11f1-bcae-25eec9cfab1b version=14f02a28-5683-3dff-9d24-b0d7956a07fe
scylladb-server | INFO 2026-03-23 12:28:40,607 [shard 0] schema_tables - Schema version changed to 47384c4-e3b5-3951-b267-25f404672fe
scylladb-server | INFO 2026-03-23 12:28:40,678 [shard 0] migration_manager - Create new ColumnFamily: org.apache.cassandra.config.CFMetadata@x600004122c00[cfId=d3090040-26b3-11f1-bcae-25eec9cfab1b, ksName=search_engine, cfName=inverted
_index, cfTypeStandard, comparator=org.apache.cassandra.db.marshall.CompositeType(org.apache.cassandra.db.marshall.Utf8Type), comment=, readRepairChance=0, dcLocalReadRepairChance=0, tombstoneOptions={mode=""}, timeout, propagation_delay_in_s
econds=""$0000}, gcTraceSeconds=864000, keyValidator=org.apache.cassandra.db.marshall.Utf8Type, minCompactionThreshold=4, maxCompactionThreshold=32, columnMetadata={ColumnDefinition(name=term, type=org.apache.cassandra.db.marshall.Utf8Type, kind=P
ARTITION_KEY, componentIndex=0, droppedAt=-9223372036854775808), ColumnDefinition(name=postings, type=org.apache.cassandra.db.marshall.Utf8Type, kind=REGULAR, componentIndex=null, droppedAt=-9223372036854775808)}, compactionStrategyCl
ass=class org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy, compactionStrategyOptions={stable_compression=org.apache.cassandra.io.compress.LZ4Compressor, bloomFilterChance=0.01, memtableFlushPeriod=0, cachingKeys="ALL", rows_per_partition="ALL", cdcId, defaultTimeToLive=0, minIndexInterval=128, maxIndexInterval=2048, speculativeRetry=99, @PERCENTILE, triggers=[]}, isDense=false, version=d89f0661-26b3-11f1-bcae-25eec9cfab1b, droppedColumns={}, collections={}, indices={}}
scylladb-server | INFO 2026-03-23 12:28:40,753 [shard 0] schema_tables - Creating search_engine.inverted_index id=d3090040-26b3-11f1-bcae-25eec9cfab1b version=ac9a770b-0cde-35ca-a0cc-3847363a2e77
scylladb-server | INFO 2026-03-23 12:28:40,873 [shard 0] schema_tables - Schema version changed to 47407d4f-402d-3a4d-bedd-8f7d6c083f5
scylladb-server | INFO 2026-03-23 12:28:40,881 [shard 0] migration_manager - Create new ColumnFamily: org.apache.cassandra.config.CFMetadata@x60000590400[cfId=d327fa10-26b3-11f1-bcae-25eec9cfab1b, ksName=search_engine, cfName=document
_stats, cfTypeStandard, comparator=org.apache.cassandra.db.marshall.CompositeType(org.apache.cassandra.db.marshall.Utf8Type), comment=, readRepairChance=0, dcLocalReadRepairChance=0, tombstoneOptions={mode=""}, timeout, propagation_delay_in_s
econds=""$0000}, gcTraceSeconds=864000, keyValidator=org.apache.cassandra.db.marshall.Utf8Type, minCompactionThreshold=4, maxCompactionThreshold=32, columnMetadata={ColumnDefinition(name=document_id, type=org.apache.cassandra.db.marshall.Utf8Type, kind=P
ARTITION_KEY, componentIndex=0, droppedAt=-9223372036854775808), ColumnDefinition(name=document_length, type=org.apache.cassandra.db.marshall.Int32Type, kind=REGULAR, componentIndex=null, droppedAt=-9223372036854775808), ColumnDefinitio
n(name=title, type=org.apache.cassandra.db.marshall.Utf8Type, kind=REGULAR, componentIndex=null, droppedAt=-9223372036854775808)}, compactionStrategyClass=class org.apache.cassandra.db.compaction.SizeTieredCompactionStrategy, compactionStr
ategyOptions={enableIndex=true, compressionParameters={stable_compression=org.apache.cassandra.io.compress.LZ4Compressor, bloomFilterChance=0.01, memtableFlushPeriod=0, cachingKeys="ALL", rows_per_partition="ALL", cdcId, defaultTimeToLive=0, minIndexInterval=128, maxIndexInterval=2048, speculativeRetry=99, @PERCENTILE, triggers=[]}, isDense=false, version=d327fa10-26b3-11f1-bcae-25eec9cfab1b, droppedColumns={}, collections={}, indices={}}
scylladb-server | INFO 2026-03-23 12:28:41,398 [shard 0] schema_tables - Creating search_engine.document_stats id=d327fa10-26b3-11f1-bcae-25eec9cfab1b version=27aa1a5d-d2eb-353f-ba5e-d3b5875f3aad
scylladb-server | INFO 2026-03-23 12:28:41,438 [shard 0] schema_tables - Schema version changed to c015631-8dd5-3e41-4630-b053ba991213
cluster-master | Schema created
cluster-master | Loading vocabulary...
cluster-master | Loaded vocabulary
cluster-master | Loading inverted index...
cluster-master | Loaded inverted index
cluster-master | Loading document stats...
cluster-master | Loaded document stats
cluster-master | Loading corpus stats...
cluster-master | Loaded corpus stats
cluster-master | All data loaded successfully
cluster-master | Store into ScyllaDB complete
cluster-master | Indexing complete
cluster-master | This script will include commands to search for documents given the query using Spark RDD
cluster-master | :: Loading settings :: url = jar:file:/usr/local/spark/jars/ivy-2.5.1.jar/org/apache/ivy/core/settings/ivysettings.xml
cluster-master | Ivy Default Cache set to: /root/.ivy2/cache
cluster-master | The jars for the packages stored in: /root/.ivy2/jars
cluster-master | com.datastax.spark#spark-cassandra-connector_2.12 added as a dependency
cluster-master | :: resolving dependencies :: org.apache.spark#spark-submit-parent-a48df7b-07a2-432f-a023-9bdc26a36471.0
cluster-master | conf: [default]
cluster-master | found com.datastax.spark#spark-cassandra-connector_2.12:3.4.0 in central
cluster-master | found com.datastax.spark#spark-cassandra-connector-driver_2.12:3.4.0 in central

```

Screenshots of successful execution of queries:

Query 1: “this is a query!”

```
-- root@vm_bd001: ~/big_data_assignment2 -- ssh root@10.90.136.133
cluster-master | 26/03/23 12:53:39 INFO BlockManagerInfo: Removed broadcast_5_piece0 on cluster-slave-1:33941 in memory (size: 9.1 KiB, free: 366.3 MiB)
cluster-master | 26/03/23 12:53:39 INFO BlockManagerInfo: Removed broadcast_5_piece0 on cluster-slave-1:46247 in memory (size: 9.1 KiB, free: 366.3 MiB)
cluster-master | 26/03/23 12:53:39 INFO SparkContext: Starting job: collect at /app/query.py:86
cluster-master | 26/03/23 12:53:39 INFO DAGScheduler: Got job 7 (collect at /app/query.py:86) with 1 output partitions
cluster-master | 26/03/23 12:53:39 INFO DAGScheduler: Final stage: ResultStage 11 (collect at /app/query.py:86)
cluster-master | 26/03/23 12:53:39 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 10)
cluster-master | 26/03/23 12:53:39 INFO DAGScheduler: Missing parents: List()
cluster-master | 26/03/23 12:53:39 INFO DAGScheduler: Submitting ResultStage 11 (MapPartitionsRDD[36] at collect at /app/query.py:86), which has no missing parents
cluster-master | 26/03/23 12:53:39 INFO BlockManagerInfo: Removed broadcast_2_piece0 on cluster-slave-1:33941 in memory (size: 5.9 KiB, free: 366.3 MiB)
cluster-master | 26/03/23 12:53:39 INFO BlockManagerInfo: Removed broadcast_2_piece0 on cluster-master:45083 in memory (size: 5.9 KiB, free: 366.2 MiB)
cluster-master | 26/03/23 12:53:39 INFO MemoryStore: Block broadcast_8 stored as values in memory (estimated size 86.7 KiB, free 366.1 MiB)
cluster-master | 26/03/23 12:53:39 INFO MemoryStore: Block broadcast_8_piece0 stored as bytes in memory (estimated size 36.7 KiB, free 366.1 MiB)
cluster-master | 26/03/23 12:53:39 INFO BlockManagerInfo: Added broadcast_8_piece0 in memory on cluster-master:45083 (size: 36.7 KiB, free: 366.2 MiB)
cluster-master | 26/03/23 12:53:39 INFO SparkContext: Created broadcast 8 from broadcast at DAGScheduler.scala:1058
cluster-master | 26/03/23 12:53:39 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 11 (MapPartitionsRDD[36] at collect at /app/query.py:86) (first 15 tasks are for partitions Vector(0))
cluster-master | 26/03/23 12:53:39 INFO YarnScheduler: Adding task set 11.0 with 1 tasks resource profile 0
cluster-master | 26/03/23 12:53:39 INFO TaskSetManager: Starting task 0.0 in stage 11.0 (TID 17) (cluster-slave-1, executor 2, partition 0, NODE_LOCAL, 9010 bytes)
cluster-master | 26/03/23 12:53:39 INFO BlockManagerInfo: Added broadcast 8_piece0 in memory on cluster-slave-1:33341 (size: 36.7 KiB, free: 366.3 MiB)
cluster-master | 26/03/23 12:53:41 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 3 to 172.18.0.3:51866
cluster-master | 26/03/23 12:53:41 INFO BlockManagerInfo: Added broadcast_7_piece0 in memory on cluster-slave-1:33341 (size: 94.4 KiB, free: 366.2 MiB)
cluster-master | 26/03/23 12:53:41 INFO TaskSetManager: Finished task 0.0 in stage 11.0 (TID 17) in 1985 ms on cluster-slave-1 (executor 2) (1/1)
cluster-master | 26/03/23 12:53:41 INFO YarnScheduler: Removed TaskSet 11.0, whose tasks have all completed, from pool
cluster-master | 26/03/23 12:53:41 INFO DAGScheduler: ResultStage 11 (collect at /app/query.py:86) Finished in 2.084 s
cluster-master | 26/03/23 12:53:41 INFO DAGScheduler: Job 7 is finished. Cancelling potential speculative or zombie tasks for this job
cluster-master | 26/03/23 12:53:41 INFO YarnScheduler: Killing all running tasks in stage 11: Stage finished
cluster-master | 26/03/23 12:53:41 INFO DAGScheduler: Job 7 finished: collect at /app/query.py:86, took 2.031004 s
cluster-master | 26/03/23 12:53:41 INFO SparkContext: SparkContext is stopping with exitCode 0.
cluster-master |
cluster-master | Search Results for: "this is a query!"
cluster-master | 1. doc_id: 42638997 - E_Coli_Metabolome_Database (score: 1.7795)
cluster-master | 2. doc_id: 47515599 - A_Canine_Sherlock_Holmes (score: 8.6785)
cluster-master | 3. doc_id: 25729657 - E-government_in_Europe (score: 2.6316)
cluster-master | 4. doc_id: 8367715 - E_verification_language (score: 1.9983)
cluster-master | 5. doc_id: 2864781 - E-folding (score: 1.9799)
cluster-master | 6. doc_id: 334877 - E-6_procs (score: 1.9237)
cluster-master | 7. doc_id: 40415365 - E-mahashadkosh (score: 1.9193)
cluster-master | 8. doc_id: 165304 - E-text (score: 1.9073)
cluster-master | 9. doc_id: 30828228 - A_Human_Right (score: 1.8823)
cluster-master | 10. doc_id: 18953915 - E_B_Williams_House (score: 1.8809)
cluster-master |
cluster-master | 26/03/23 12:53:42 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
cluster-master | 26/03/23 12:53:42 INFO YarnClientSchedulerBackend: Interrupting monitor thread
cluster-master | 26/03/23 12:53:42 INFO YarnClientSchedulerBackend: Shutting down all executors
cluster-master | 26/03/23 12:53:42 INFO YarnSchedulerBackend$YarnDriverEndpoint: Asking each executor to shut down
cluster-master | 26/03/23 12:53:42 INFO YarnClientSchedulerBackend: YARN client scheduler backend Stopped
cluster-master | 26/03/23 12:53:42 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
cluster-master | 26/03/23 12:53:42 INFO BlockManager: MemoryStore cleared
cluster-master | 26/03/23 12:53:42 INFO BlockManager: BlockManager stopped
cluster-master | 26/03/23 12:53:42 INFO BlockManagerMaster: BlockManagerMaster stopped
cluster-master | 26/03/23 12:53:42 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
cluster-master | 26/03/23 12:53:42 INFO SparkContext: Successfully stopped SparkContext
cluster-master | 26/03/23 12:53:43 INFO ShutdownHookManager: Shutdown hook called
cluster-master | 26/03/23 12:53:43 INFO ShutdownHookManager: Deleting directory /tmp/spark-758f491e-bddf-4cf3-bf69-96dc8b3363bc
cluster-master | 26/03/23 12:53:43 INFO ShutdownHookManager: Deleting directory /tmp/spark-1c0432bd-fd92-413d-baa7-73689d5d9ae1/pyspark-c077bd7c-a42d-4811-a009-ccccf65671c5
cluster-master | 26/03/23 12:53:43 INFO ShutdownHookManager: Deleting directory /tmp/spark-1c0432bd-fd92-413d-baa7-73689d5d9ae1
cluster-master | 26/03/23 12:53:43 INFO CassandraConnector: Disconnected from Cassandra cluster.
cluster-master | 26/03/23 12:53:43 INFO SerialShutdownHooks: Successfully executed shutdown hook: Clearing session cache for C* connector
cluster-master | cluster-master exited with code 0

```

Query 2: “Hello, World!”


```
-- root@vm_bd001: ~big_data_assignment2 -- ssh root@10.90.136.133 -- root@cluster-master: /app -- ssh root@10.90.136.133 +

26/03/23 13:00:18 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but current version of codegened fast hashmap does not support this aggregate.
26/03/23 13:00:18 INFO CodeGenerator: Code generated in 128.434202 ms
26/03/23 13:00:18 INFO SparkContext: Starting job: collect at /app/query.py:86
26/03/23 13:00:18 INFO DAGScheduler: Got job 7 (collect at /app/query.py:86) with 1 output partitions
26/03/23 13:00:18 INFO DAGScheduler: Final stage: ResultStage 11 (collect at /app/query.py:86)
26/03/23 13:00:18 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 10)
26/03/23 13:00:18 INFO DAGScheduler: Missing parents: List()
26/03/23 13:00:18 INFO DAGScheduler: Submitting ResultStage 11 (MapPartitionsRDD[36] at collect at /app/query.py:86), which has no missing parents
26/03/23 13:00:18 INFO MemoryStore: Block broadcast_8 stored as values in memory (estimated size 86.7 KiB, free 366.0 MiB)
26/03/23 13:00:18 INFO MemoryStore: Block broadcast_8_piece0 stored as bytes in memory (estimated size 36.8 KiB, free 366.0 MiB)
26/03/23 13:00:18 INFO BlockManagerInfo: Added broadcast_8_piece0 in memory on cluster-master:34087 (size: 36.8 KiB, free: 366.2 MiB)
26/03/23 13:00:18 INFO SparkContext: Created broadcast 8 from broadcast at DAGScheduler.scala:1585
26/03/23 13:00:18 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 11 (MapPartitionsRDD[36] at collect at /app/query.py:86) (first 15 tasks are for partitions Vector(0))
26/03/23 13:00:18 INFO YarnScheduler: Adding task set 11.0 with 1 tasks resource profile 0
26/03/23 13:00:18 INFO TaskSetManager: Starting task 0.0 in stage 11.0 (TID 17) (cluster-slave-1, executor 1, partition 0, NODE_LOCAL, 9010 bytes)
26/03/23 13:00:18 INFO BlockManagerInfo: Added broadcast_8_piece0 in memory on cluster-slave-1:34309 (size: 36.8 KiB, free: 366.2 MiB)
26/03/23 13:00:19 INFO MapOutputTrackerMasterEndpoint: Asked to send map output locations for shuffle 3 to 172.18.0.3:56668
26/03/23 13:00:20 INFO BlockManagerInfo: Added broadcast_7_piece0 in memory on cluster-slave-1:34309 (size: 10.0 KiB, free: 366.2 MiB)
26/03/23 13:00:20 INFO YarnScheduler: Finished task 0.0 in stage 11.0 (TID 17) in 1051 ms on cluster-slave-1 (executor 1) (1/1)
26/03/23 13:00:20 INFO YarnScheduler: Removed TaskSet 11.0, whose tasks have all completed, from pool
26/03/23 13:00:20 INFO DAGScheduler: ResultStage 11 (collect at /app/query.py:86) finished in 1.662 s
26/03/23 13:00:20 INFO DAGScheduler: Job 7 is finished. Cancelling potential speculative or zombie tasks for this job
26/03/23 13:00:20 INFO YarnScheduler: Killing all running tasks in stage 11: Stage finished
26/03/23 13:00:20 INFO DAGScheduler: Job 7 finished: collect at /app/query.py:86, took 1.684852 s

Search Results for: "Hello, World!"
1. doc_id: 64251145 - A_Couple_of_Cuckoos (score: 8.7122)
2. doc_id: 4725243 - E_(Australia_and_New_Zealand) (score: 7.9221)
3. doc_id: 53756721 - A_Doll's_House_Part_2 (score: 6.9636)
4. doc_id: 2138506 - E-Z_Listening_Disc (score: 6.7536)
5. doc_id: 31671818 - E_mu_SP-12 (score: 6.5693)
6. doc_id: 3217028 - A_complicated_Kindness (score: 4.9798)
7. doc_id: 5517618 - A_Boogie_wit_da_Hoodie_discography (score: 4.4670)
8. doc_id: 36993393 - E-girls (score: 2.7636)
9. doc_id: 52462115 - E_Bower_Carty (score: 2.7361)
10. doc_id: 32831169 - E_Alexander_Powell (score: 2.7058)

26/03/23 13:00:20 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/03/23 13:00:20 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
26/03/23 13:00:20 INFO YarnClientSchedulerBackend: Interrupting monitor thread
26/03/23 13:00:20 INFO YarnClientSchedulerBackend: Shutting down all executors
26/03/23 13:00:20 INFO YarnSchedulerBackendYarnDriverEndpoint: Asking each executor to shut down
26/03/23 13:00:20 INFO YarnClientSchedulerBackend: YARN client scheduler backend Stopped
26/03/23 13:00:20 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/03/23 13:00:21 INFO MemoryStore: MemoryStore cleared
26/03/23 13:00:21 INFO BlockManager: BlockManager stopped
26/03/23 13:00:21 INFO BlockManagerMaster: BlockManagerMaster stopped
26/03/23 13:00:21 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/03/23 13:00:21 INFO SparkContext: Successfully stopped SparkContext
26/03/23 13:00:21 INFO ShutdownHookManager: Shutdown hook called
26/03/23 13:00:21 INFO ShutdownHookManager: Deleting directory /tmp/spark-069b478f-8939-4337-8c1f-1117e46a6782
26/03/23 13:00:21 INFO ShutdownHookManager: Deleting directory /tmp/spark-b38384fe-810b-4e5b-a2e5-ea898468159f/pyspark-2bd8bb27-9a28-48fa-9b65-b084f39ea430
26/03/23 13:00:21 INFO ShutdownHookManager: Deleting directory /tmp/spark-b38384fe-810b-4e5b-a2e5-ea898468159f
26/03/23 13:00:22 INFO CassandraConnector: Disconnected from Cassandra cluster.
26/03/23 13:00:22 INFO SerialShutdownHooks: Successfully executed shutdown hook: Clearing session cache for C* connector
root@cluster-master:/app
```

Query 3: “cats and dogs”

```
-- root@vm_bd001: ~big_data_assignment2 -- ssh root@10.90.136.133 -- root@cluster-master: /app -- ssh root@10.90.136.133 +

26/03/23 13:03:56 INFO HashAggregateExec: spark.sql.codegen.aggregate.map.twolevel.enabled is set to true, but current version of codegened fast hashmap does not support this aggregate.
26/03/23 13:03:56 INFO CodeGenerator: Code generated in 145.148432791 ms
26/03/23 13:03:57 INFO SparkContext: Starting job: collect at /app/query.py:86
26/03/23 13:03:57 INFO DAGScheduler: Got job 7 (collect at /app/query.py:86) with 1 output partitions
26/03/23 13:03:57 INFO DAGScheduler: Final stage: ResultStage 11 (collect at /app/query.py:86)
26/03/23 13:03:57 INFO DAGScheduler: Parents of final stage: List(ShuffleMapStage 10)
26/03/23 13:03:57 INFO DAGScheduler: Missing parents: List()
26/03/23 13:03:57 INFO DAGScheduler: Submitting ResultStage 11 (MapPartitionsRDD[36] at collect at /app/query.py:86), which has no missing parents
26/03/23 13:03:57 INFO MemoryStore: Block broadcast_8 stored as values in memory (estimated size 86.7 KiB, free 366.0 MiB)
26/03/23 13:03:57 INFO MemoryStore: Block broadcast_8_piece0 stored as bytes in memory (estimated size 36.8 KiB, free 366.0 MiB)
26/03/23 13:03:57 INFO BlockManagerInfo: Added broadcast_8_piece0 in memory on cluster-master:38117 (size: 36.8 KiB, free: 366.2 MiB)
26/03/23 13:03:57 INFO SparkContext: Created broadcast 8 from broadcast at DAGScheduler.scala:1585
26/03/23 13:03:57 INFO DAGScheduler: Submitting 1 missing tasks from ResultStage 11 (MapPartitionsRDD[36] at collect at /app/query.py:86) (first 15 tasks are for partitions Vector(0))
26/03/23 13:03:57 INFO YarnScheduler: Adding task set 11.0 with 1 tasks resource profile 0
26/03/23 13:03:57 INFO TaskSetManager: Starting task 0.0 in stage 11.0 (TID 17) (cluster-slave-1, executor 1, partition 0, NODE_LOCAL, 9010 bytes)
26/03/23 13:03:57 INFO BlockManagerInfo: Added broadcast_8_piece0 in memory on cluster-slave-1:45223 (size: 36.8 KiB, free: 366.2 MiB)
26/03/23 13:03:57 INFO MemoryStore: Block broadcast_8_piece0 stored as bytes in memory (estimated size 36.8 KiB, free 366.0 MiB)
26/03/23 13:03:57 INFO BlockManagerInfo: Added broadcast_7_piece0 in memory on cluster-slave-1:45223 (size: 41.0 KiB, free: 366.2 MiB)
26/03/23 13:03:57 INFO TaskSetManager: Finished task 0.0 in stage 11.0 (TID 17) in 1471 ms on cluster-slave-1 (executor 1) (1/1)
26/03/23 13:03:58 INFO YarnScheduler: Removed TaskSet 11.0, whose tasks have all completed, from pool
26/03/23 13:03:58 INFO DAGScheduler: ResultStage 11 (collect at /app/query.py:86) finished in 1.481 s
26/03/23 13:03:58 INFO DAGScheduler: Job 7 is finished. Cancelling potential speculative or zombie tasks for this job
26/03/23 13:03:58 INFO YarnScheduler: Killing all running tasks in stage 11: Stage finished
26/03/23 13:03:58 INFO DAGScheduler: Job 7 finished: collect at /app/query.py:86, took 1.496234 s

Search Results for: "cats and dogs"
1. doc_id: 10858097 - A_Dangerous_Path (score: 14.6933)
2. doc_id: 35739004 - A_Dog's_Purpose (score: 8.4423)
3. doc_id: 18548314 - A_Cat_Abroad (score: 6.8990)
4. doc_id: 42835467 - A_Cat_a_Mouse_and_a_Bell (score: 6.8892)
5. doc_id: 64378948 - A_Gift_from_Bob (score: 6.6131)
6. doc_id: 6877746 - A_Land_Remembered (score: 5.9602)
7. doc_id: 48516581 - A_Dog_Named_Gucci (score: 5.9492)
8. doc_id: 7879205 - A_Dog_Year (score: 5.8781)
9. doc_id: 48232752 - A_Little_Is_Enough (score: 5.7789)
10. doc_id: 17488265 - A_Boy_and_His_Dog_(1946_film) (score: 5.7586)

26/03/23 13:03:58 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/03/23 13:03:58 INFO SparkUI: Stopped Spark web UI at http://cluster-master:4040
26/03/23 13:03:58 INFO YarnClientSchedulerBackend: Interrupting monitor thread
26/03/23 13:03:59 INFO YarnClientSchedulerBackend: Shutting down all executors
26/03/23 13:03:59 INFO YarnSchedulerBackendYarnDriverEndpoint: Asking each executor to shut down
26/03/23 13:03:59 INFO YarnClientSchedulerBackend: YARN client scheduler backend Stopped
26/03/23 13:03:59 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/03/23 13:03:59 INFO MemoryStore: MemoryStore cleared
26/03/23 13:03:59 INFO BlockManager: BlockManager stopped
26/03/23 13:03:59 INFO BlockManagerMaster: BlockManagerMaster stopped
26/03/23 13:03:59 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
26/03/23 13:03:59 INFO SparkContext: Successfully stopped SparkContext
26/03/23 13:03:59 INFO ShutdownHookManager: Shutdown hook called
26/03/23 13:03:59 INFO ShutdownHookManager: Deleting directory /tmp/spark-6d282252-7025-4d28-bb87-f847125a31d1/pyspark-a4b1b560-25ab-485c-98fe-d1382181dd4f
26/03/23 13:03:59 INFO ShutdownHookManager: Deleting directory /tmp/spark-6d282252-7025-4d28-bb87-f847125a31d1
26/03/23 13:03:59 INFO ShutdownHookManager: Deleting directory /tmp/spark-5df2e68c-18d3-4c50-b87c-be5f0cdeb17b
26/03/23 13:04:00 INFO CassandraConnector: Disconnected from Cassandra cluster.
26/03/23 13:04:00 INFO SerialShutdownHooks: Successfully executed shutdown hook: Clearing session cache for C* connector
root@cluster-master:/app
```

Explanations of results and reflections on findings:

Although I used e.parquet as the input data, some search results returned articles starting with "A". This occurred because when I cloned the repository on the virtual machine (which I used because YARN worked there, unlike on my laptop), I did not delete the existing data folder that came with the repository template. This folder already contained articles beginning with "A". Since the /app folder is mounted in

Docker, when I ran `prepare_data.py`, the newly processed articles from `e.parquet` were added to the local `/app/data` folder, which already held the older "A" articles. Then, when `prepare_data.sh` copied everything from `/app/data` to HDFS, both sets of articles were uploaded. As a result, the index contained both the intended "E" articles and the leftover "A" articles.

The first query returned "`E._Coli_Metabolome_Database`" (score: 11.7795) as the top result, which is believable for a document about a database containing the word "query". The second result was "`A_Canine_Sherlock_Holmes`" (score: 8.8785), which could also contain the word "query" given its connection to detective stories. The remaining documents received much lower scores (ranging from 2.6316 to 1.8809), likely because they only contained common stop words like "this", "is", and "a" without matching any meaningful query terms.

The second query returned "`A_Couple_of_Cuckoos`" (score: 8.7122) as the top result because the document mentions the song "HELLO HELLO HELLO" twice. The other results have scores ranging from 7.9221 to 2.7058. This score distribution is smoother than the previous query because both query words are fairly common, yet not stop words, so the scores decrease more smoothly from the top result to the bottom.

The third query returned "`A_Dangerous_Path`" (score: 14.6933) as the top result, because its text contains both "cats" and "dogs" with high frequency. The next two results, "`A_Dog's_Purpose`" (score: 8.4423) and "`A_Cat_Abroad`" (score: 6.8990), clearly match one query term in their titles. The remaining results show scores between 6.89 and 5.76, with several titles containing "dog" ("`A_Dog_Named_Gucci`", "`A_Dog_Year`", "`A_Boy_and_His_Dog`") and others likely containing the terms in their text with lower frequency. This query produced the highest overall scores, showing that BM25 gives higher scores to documents that contain all the query words, and lower scores to those that contain only some of them.

Based on queries' results, the search engine works and successfully retrieves relevant documents. The BM25 implementation correctly gives higher scores to documents that match query terms and lower scores to those with only partial matches. However, generic queries like "this is a query!" produced low scores because stop words ("this", "is", "a") dominated the search. Adding stop word removal could improve precision for such queries by ignoring common words that carry little meaning. Also, a title boost could help by giving extra weight to terms appearing in document titles, which often indicate strong relevance.